

BINAY—Lighting Up The Future

Binay LED Aviation Obstruction Lights



**BINAY LED-based MEDIUM INTENSITY
Aviation Obstruction Light Beacon**

As per the International Civil Aviation Organisation (ICAO) requirements for use on structures above 45m

Available in Low Intensity and Medium Intensity versions (as per the International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with a 5-year warranty

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave, and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



**THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED
PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT**



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prototype that can connect two communicating nodes through fibre-optic cable, allowing more secure and less vulnerable information transfer.

The elementary quantum network apparently consists of two coupled single-atom nodes that communicate quantum information via coherent exchange of single photons. Quantum computing works on qubits, which specify the spin or orientation of an atom. The problem is they can't be cloned like binary bits, which brings in the need of a better and sophisticated mechanism for their transfer.

Researchers at MPQ tried to achieve the same by creating a set-up that used two nodes connected with each other through an optic fibre cable and a single rubidium atom trapped in a reflective optical cavity.

On the flip side, several problems are associated with the technique. The atoms are highly sensitive to the temperature, which can give unexpected results. Also, the accuracy of the entire process of transferring the information is 80 to 85 per cent.

"Entanglement of two systems separated by a large distance is a fascinating phenomenon in itself. However, it could also serve as a resource for teleportation of quantum information. One day, this might not only make it possible to communicate quantum information over very large distances but also enable an entire quantum Internet," Stephan Ritter, a researcher at MPQ, said.

Multicore chips to improve your PC's speed

Future computer chips are expected to have hundreds or even thousands of cores. However, only a pair of cores can communicate over a bus at present.

Li-Shiuan Peh, associate professor of electrical engineering and computer science at MIT, and his colleagues aim to address this issue by bundling the information carried by these cores into packets so that they can communicate in the same way as networked computers do. Supposedly, each core would have its own router responsible for sending the packets over the network. Multiple cores are faster as these can run multiple instructions at the same time, increasing the overall speed for programs.

The virtual bypassing technique developed by the team facilitates the router to send an advance signal to the next. This helps the latter to preset its switch, increasing the speed of the packet.

"Virtual bypassing allowed a very close approach to the maximum data-transmission rates predicted by theoretical analysis," Peh added.

These findings will soon be presented in *Design Automation Conference* in the US.

Baby brain power being tapped for smarter computers

Computational models based on brainpower of babies, toddlers and preschoolers could make artificial intelligence smarter by overcoming limitations such as handling nuances and uncertainty—according to researchers. In a wide range of experiments involving lollipops, flashing and spinning toys, and music makers, researchers have found that